

* Note: Payloads are shown in JSON/Plain-text format for clarity, but are transmitted as packed C-structs over LoRa to minimize time-on-air, battery usage, and collision probability.

Node (Mota)
Client

Router
Coordinator

Backend
(via SIM800L)

User selects Network
State: STATE_START_JOIN
IsChannelFree() Check

JOIN_REQUEST (0x02)

Payload: [Router: 10, ID: 50, fcnt = ++join_cnt]

Anti-Replay Validation
(lastJoinCnt < incomingJoinCnt)

ALT: Public Network (Immediate Decision)

Check: *connectedClients* < *MAX_CLIENTS*

JOIN_ACCEPTED (0x03) / JOIN_DENIED (0x04)

Payload: ControlData

ALT: Private Network (Async Backend Verification)

enqueueAccessCheck(CLIENT_ID)
Wait for Backend

HTTP POST (AM-036)

Payload: {"id.mota": 50, "routerApiToken": "xyz..."}

HTTP 200 OK

Validation Result

JOIN_ACCEPTED (0x03) / JOIN_DENIED (0x04)

Payload: ControlData

If Accepted: STATE_START_DATA
If Denied / Timeout: STATE_SLEEP